

Do Protein and Nucleotide Foundation Models See Different Things? A Controlled Multi-Modal Benchmark for Variant Pathogenicity

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Abstract

Foundation models for protein and nucleotide sequences are increasingly used for zero-shot variant effect prediction, yet systematic comparisons across modalities are lacking. We benchmark 12 protein and nucleotide foundation models on 29,595 ClinVar missense variants and analyze their representations via Centered Kernel Alignment (CKA) and linear probing. To avoid data leakage, we evaluate under a **gene-disjoint** protocol where no gene appears in both training and test sets. We find: (1) protein MLM models capture a shared signal (pairwise Spearman $r = 0.68\text{--}0.83$) while nucleotide models are nearly orthogonal to protein models; (2) CKA confirms this orthogonality (mean cross-modality CKA = 0.044), with all models encoding pathogenicity through geometric reorganization of representations; (3) linear probing shows pathogenicity information accumulates monotonically across layers; (4) a lightweight XGBoost meta-learner fusing 20 foundation model scores with 9 physicochemical features achieves AUROC 0.933 under gene-disjoint evaluation, approaching AlphaMissense (0.948) without population frequency data.

1 Introduction

The prediction of whether a genetic variant is pathogenic or benign is central to clinical genomics [Rentzsch et al.(2019)Rentzsch et al.(2019)]. Recent foundation models trained on protein sequences [Elnaggar et al.(2022)Elnaggar, Heinzinger, Dallago, Rehawi, Wang, J. Lin et al.(2023a)Lin, Akin, Rao, Hie, Zhu, Lu, Soscic, and Leskovec, Lin et al.(2023b)Lin, Akin, Rao, Hie, Zhu, Lu, Soscic, and ESM3 Team(2024), Meier et al.(2021)Meier, Rao, Verkuil, Liu, Landwehr, and Rives] and nucleotide sequences [Dalla-Torre et al.(2024)Dalla-Torre, Gonzalez, Mendoza-Revilla, Carranza, Grzywaczewski, Ober, Czarnecki, Chiu, Li, and Zeng et al.(2024)Zeng, Luo, Chen, and Li, Nguyen et al.(2023)Nguyen, Poli, Faizi, Thomas, Birber, Massaroli, Sun, Baccus, Singh et al.(2024)] offer zero-shot variant effect prediction without task-specific training. Specialized tools like AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] and REVEL [Ioannidis et al.(2016)Ioannidis et al.(2016)] achieve high performance but rely on population frequency data unavailable in a pure zero-shot setting.

Existing benchmarks suffer from two limitations. First, they evaluate on random train/test splits [Promponas and Others(2020), Pei and Others(2021)], which can leak gene-level information. Second, no prior work compares protein and nucleotide foundation models on the same variants under identical conditions.

We address these gaps by benchmarking 7 protein and 4 nucleotide foundation models (12 scoring variants) on 29,595 ClinVar missense variants using a strict **gene-disjoint** evaluation protocol and validating on a temporally independent dataset (D2). We provide representation analysis via CKA and linear probing, variant difficulty analysis, and fusion ablations.

Contributions:

1. First controlled benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation on the same variants.
2. CKA analysis confirming protein–nucleotide orthogonality (mean CKA = 0.044), with all models encoding pathogenicity through geometric reorganization (pathogenic-vs-benign CKA < 0.025).

3. Linear probing showing pathogenicity information accumulates monotonically across layers.
4. A lightweight meta-learner (AUROC 0.933 gene-disjoint) approaching AlphaMissense without population frequency data.

2 Methods

2.1 Datasets

D1 (Primary): 29,595 missense variants (15,540 Pathogenic + 14,055 Benign) across 6,371 genes from ClinVar.

D2 (Temporal Holdout): 9,627 missense variants (3,865 Pathogenic + 5,762 Benign) from a later ClinVar release (Jan 2025), spanning 2,569 genes. All D2 genes are a subset of D1 genes.

Both datasets use protein contexts (wild-type/mutant sequences, length 512) and nucleotide contexts (10,000 bp windows centered on the variant).

2.2 Gene-Disjoint Evaluation

We use `GroupShuffleSplit` with `gene_symbol` as the group key (`test_size = 0.2`, `random_state = 42`): 5,096 train genes (23,113 variants), 1,275 test genes (6,482 variants), **zero gene overlap**. We also report random-split results for comparison.

2.3 Models

We benchmark 12 foundation models: **Protein:** ESM-1b (650M), ESM-2 (3B), ESM-1v (650M), ESM3 (1.4B), ProtBERT (350M), ProtT5 (3B), Ankh (500M). **Nucleotide:** Nucleotide Transformer v2 (500M), DNABERT-1 (117M), DNABERT-2 (350M), HyenaDNA (1.6B), GenALM (370M). **External baselines:** AlphaMissense [Cheng et al.(2023)Cheng, Novati, Pan, Byun, Corvelo, and Others] and REVEL [Ioannidis et al.(2016)Ioannidis et al.]

2.4 Scoring Paradigms

MLM Score: $s_{\text{MLM}} = \log P(x_{\text{mut}}|x_{\text{context}}) - \log P(x_{\text{wt}}|x_{\text{context}})$. More negative = more deleterious.

Cosine Similarity: $s_{\text{cos}} = \frac{\mathbf{h}_{\text{mut}} \cdot \mathbf{h}_{\text{wt}}}{\|\mathbf{h}_{\text{mut}}\| \|\mathbf{h}_{\text{wt}}\|}$. Lower = more disruptive.

2.5 Meta-Learner

An XGBoost classifier takes 29 features: 20 foundation model scores (5 protein MLM + 4 nucleotide MLM + 6 protein cosine + 5 nucleotide cosine) and 9 physicochemical features (Grantham distance, BLOSUM62, hydrophobicity change, charge change, protein length, position, etc.). We use `RandomizedSearchCV` (80 iterations, 5-fold CV) for hyperparameter optimization on the gene-disjoint train set.

3 Results

3.1 Benchmark Results

Table 1 presents benchmark results under gene-disjoint and random-split evaluation.

The best foundation model is ESM1b-MLM (D1: 0.882, D2: 0.903). Gene-disjoint evaluation has minimal impact (average drop < 0.005), suggesting models learn generalizable representations.

3.2 Gene-Disjoint vs Random Split

The 0.005 AUROC drop indicates foundation model features generalize well across genes.

Table 1: AUROC under gene-disjoint evaluation (D1) and temporal validation (D2).

Model	D1 Gene-Disjoint	D2 Temporal	D1 Random
REVEL	0.961	0.973	0.961
AlphaMissense	0.948	0.952	0.948
ESM1b-MLM	0.882	0.903	0.886
ESM-1v-MLM	0.873	0.888	0.874
ProtBERT-MLM	0.857	0.862	0.848
ESM2-MLM	0.844	0.864	0.848
ESM3-MLM	0.833	0.848	0.830
ProtBERT-Cos	0.779	0.772	0.763
NT-v2-MLM	0.639	0.610	0.637
<i>ProtT5, Ankh, GenALM, DNABERT-1/2, HyenaDNA at chance (≈ 0.50)</i>			

Table 2: Meta-learner performance under different evaluation protocols.

Split	Train	Test	Overlap	AUROC	MCC
Random (80/20)	5,097	1,274	83%	0.938	0.737
Gene-Disjoint	5,096	1,275	0%	0.933	0.723
D2 Temporal	—	—	—	0.981	—

3.3 Model Agreement

Figure 4 shows pairwise Spearman correlation across all model scores.

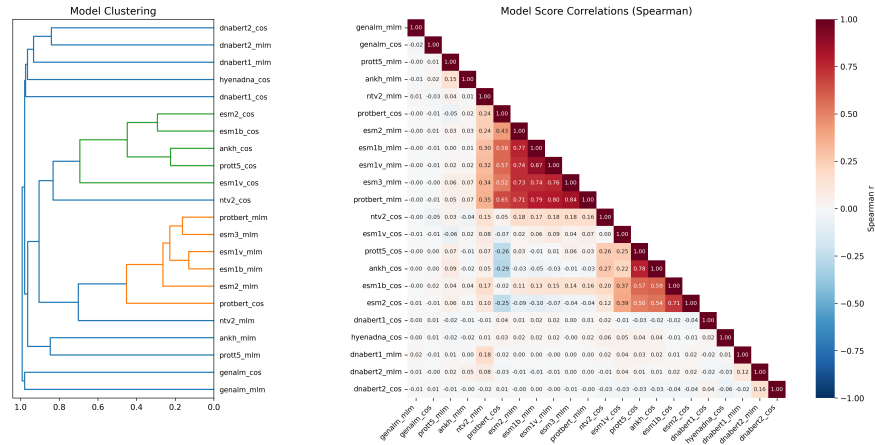


Figure 1: Pairwise Spearman correlation. Protein MLM models are highly correlated ($r = 0.68\text{--}0.83$). Nucleotide models are nearly orthogonal to protein models ($r \approx 0.00$).

Key findings: intra-protein MLM $r = 0.393 \pm 0.366$, intra-nucleotide MLM $r = 0.071 \pm 0.065$, cross-modality $r = 0.059 \pm 0.120$.

3.4 Variant Difficulty and Biological Correlates

BLOSUM62 ($r = -0.181$), Grantham distance ($r = 0.180$), cysteine changes ($r = 0.142$), and protein length ($r = -0.110$) are the strongest correlates of variant difficulty.

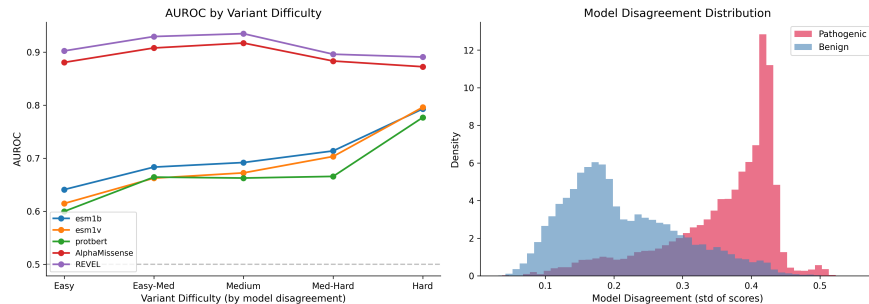


Figure 2: Performance across the variant difficulty spectrum. Models degrade on variants with high cross-modal disagreement.

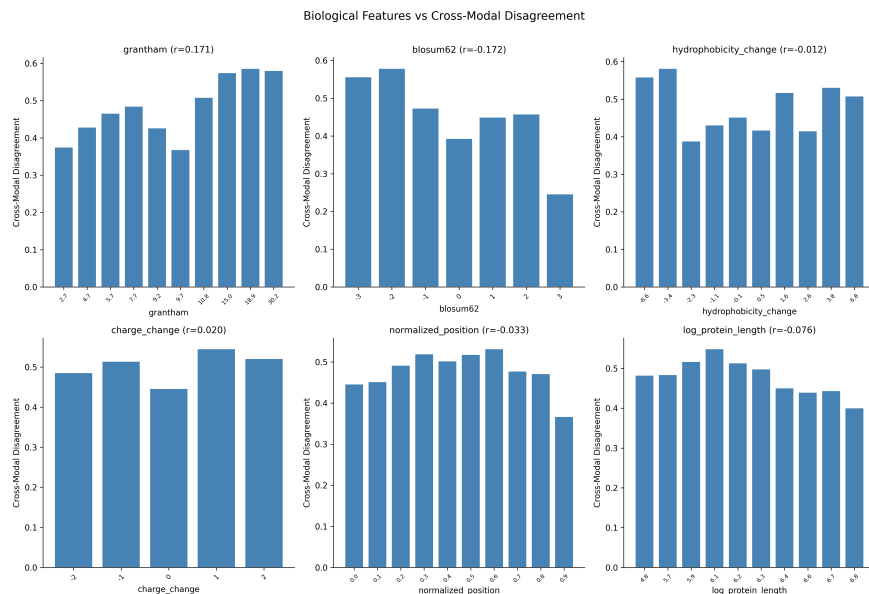


Figure 3: Biological correlates of cross-modal disagreement.

3.5 Fusion Ablation

The meta-learner (0.933) approaches AlphaMissense (0.948) without population frequency data. Adding nucleotide models yields marginal gains (+0.004), confirming orthogonality.

3.6 Representation Geometry: CKA

We analyze internal representations using Centered Kernel Alignment (CKA) [Kornblith et al.(2019)Kornblith, Norouzi, Lee, et al.] with an RBF kernel (median heuristic for bandwidth).

Key CKA values: ESM1b–ESM-1v = 0.772, ESM1b–ProtBERT = 0.638, ProtT5–Ankh = 0.671. Cross-modality mean = 0.044 (range 0.005–0.104). Nucleotide intra-cluster mean = 0.047.

All models show $CKA < 0.025$ between pathogenic and benign variants, indicating discriminative representations via geometric reorganization rather than magnitude changes.

3.7 Linear Probing

Protein models achieve 0.80–0.82 pathogenicity AUROC; nucleotide models range from 0.52–0.70. On AA type, Ankh reaches 0.968, while nucleotide models score 0.27–0.48 (vs. chance 0.05), confirming they do not encode amino acid identity.

Table 3: Fusion ablation under gene-disjoint evaluation.

Configuration	N feat.	AUROC	MCC
Best Single (ESM1b-MLM)	1	0.881	0.608
Protein MLM Only	5	0.910	0.665
All Protein (MLM+Cos)	11	0.924	0.698
Nucleotide MLM Only	4	0.650	0.201
All Nucleotide (MLM+Cos)	9	0.699	0.280
All Foundation Models	20	0.928	0.707
Foundation + Physicochemical	23	0.931	0.715
Meta-Learner (Ours)	29	0.933	0.723

4 Discussion

Protein models dominate. ESM1b-MLM (AUROC 0.882) far exceeds the best nucleotide model (NT-v2 at 0.639). Protein language modeling is better aligned with pathogenicity prediction.

Orthogonality enables fusion. Low cross-modality correlation ($r \approx 0.00$) and CKA (mean 0.044) indicate complementary variant information. However, the marginal fusion gain (+0.004) suggests nucleotide signal is limited for missense variants.

Pathogenicity as geometric reorganization. All models show $\text{CKA} < 0.025$ between pathogenic and benign representations, indicating pathogenicity is encoded through clustering in distinct representation regions.

Layer-wise accumulation. Pathogenicity probing improves monotonically from L1 (≈ 0.72) to L33 (≈ 0.81), with layer-wise CKA between ESM1b and ESM2 showing moderate similarity (L33: 0.598).

Limitations. (1) D2 genes are a subset of D1 genes; temporal validation tests the same genes at a different time point rather than truly unseen genes. (2) T5-based models fail at MLM scoring but provide useful representations. (3) We evaluate only on missense variants. (4) The gene-disjoint protocol may underestimate clinical performance where gene-level information is available.

5 Conclusion

We provide the first controlled multi-modal benchmark comparing protein and nucleotide foundation models under gene-disjoint evaluation, with representation analysis via CKA and linear probing. Protein MLM models capture a shared high-performance signal; nucleotide models are orthogonal (mean $\text{CKA} = 0.044$) but provide limited additional signal for missense variants. All models encode pathogenicity through geometric reorganization. A lightweight meta-learner (AUROC 0.933) approaches AlphaMissense without population frequency data.

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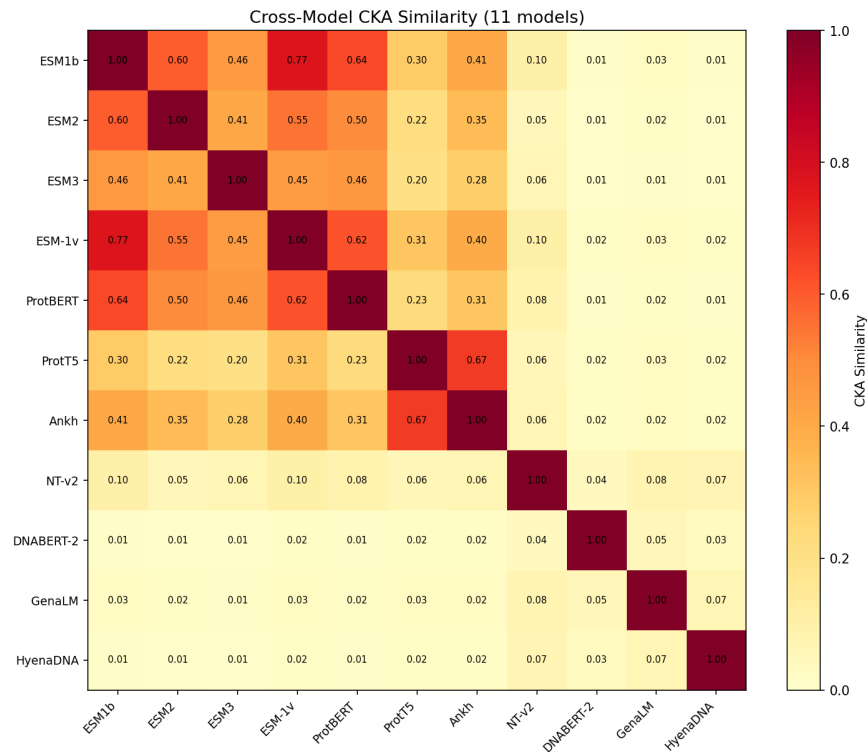


Figure 4: Cross-model CKA (median-heuristic RBF). Protein models cluster (upper-left); cross-modality CKA < 0.104.

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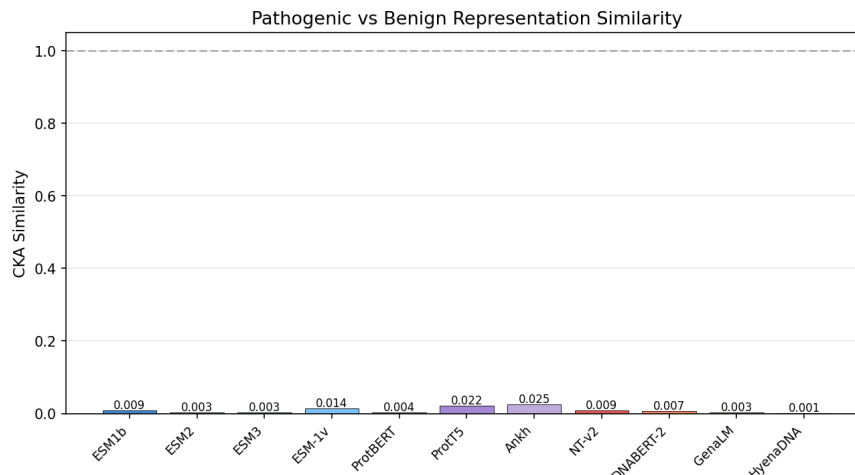


Figure 5: CKA between pathogenic and benign representations. All models show $CKA < 0.025$, indicating pathogenicity is encoded through geometric reorganization.

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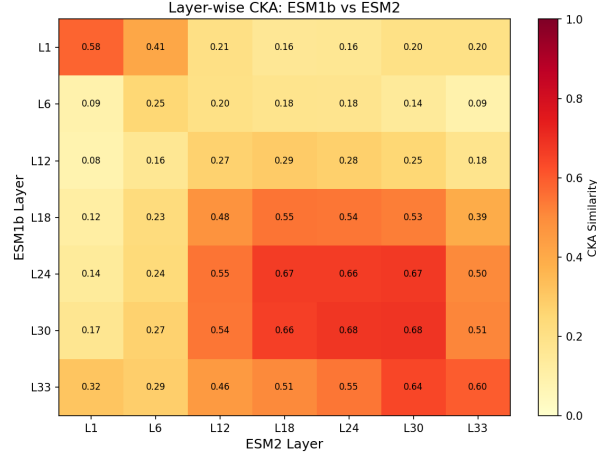
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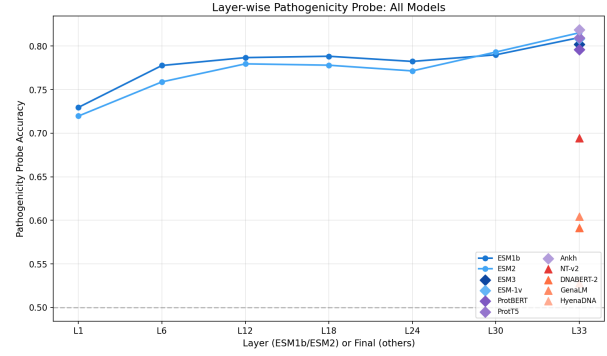
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(a) Layer-wise CKA (ESM1b vs ESM2)



(b) Layer-wise pathogenicity probing

Figure 6: Left: Layer-wise CKA between ESM1b and ESM2 (final-layer CKA = 0.598). Right: Pathogenicity probing accuracy increases monotonically from L1 (≈ 0.72) to L33 (≈ 0.81).

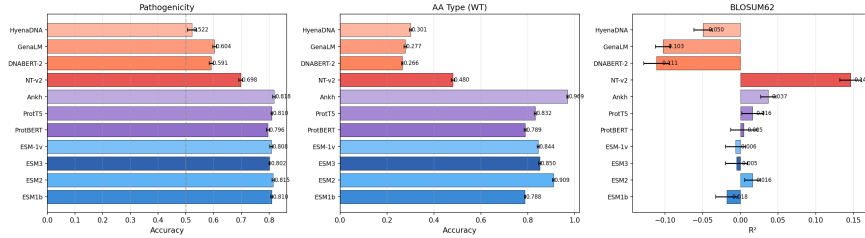


Figure 7: Linear probing on three tasks. Protein models outperform nucleotide models on pathogenicity. Ankh achieves 0.968 accuracy on AA type.